

Industrial Automated Sorting and Material Handling Cell

CODESYS LD + Factory I/O + Modbus TCP

Industrial Automation Portfolio
Project Design Report

Abstract—This project specifies a virtual material handling cell that sorts boxes by height using conveyor transport and a turntable. CODESYS Ladder Diagram provides the primary control logic, Factory I/O represents the plant, and Modbus TCP exchanges sensor and actuator data. The design combines an explicit operating sequence, automatic and manual modes, motion interlocks, latched diagnostics, and production counters. Low boxes are assigned to the left conveyor and high boxes to the right. This report describes the intended architecture and acceptance criteria. No controller runtime logs or completed test records were available for this report; therefore, integration success, routing accuracy, cycle time, and fault response remain unverified.

Index Terms—PLC, Ladder Diagram, CODESYS, Factory I/O, Modbus TCP, material handling

I. Objective and Scope

The objective is a maintainable industrial automation portfolio project demonstrating discrete sequencing, diagnostics, and reproducible commissioning. The selected virtual plant is Factory I/O's Sorting by Height (Advanced), whose documented objective is height sorting with a turntable [1]. The project routing requirement is low-to-left and high-to-right. This is a design report, not evidence of a commissioned machine or a validated controller implementation.

II. System Architecture

The proposed architecture has three layers: the simulated plant, a cyclic PLC application, and operator/status variables. Factory I/O provides box detection, height sensing, turntable position feedback, and branch occupancy signals. CODESYS receives these inputs, evaluates mode and fault conditions, advances the sequence, and writes gated actuator commands. The operator layer exposes Start, Stop, Reset, Auto/Manual selection, jog requests, state, alarms, and counters.

The PLC structure separates plant I/O variables, operator/status variables, sequence logic, and final output assignments. Each actuator has a single final command assignment to avoid conflicting writes. Input polarity and scene tag semantics must be documented before commissioning. A conservative one-box-at-a-time admission policy keeps the latched class associated with its box; overlapping transport requires additional tracking and is outside this baseline.

III. Control Strategy

A. Ladder Logic and Operating Sequence

Ladder Diagram (LD) is the primary implementation language. Contacts and coils express run permissives and interlocks; edge detectors, on-delay timers, and counters support event handling. The normal sequence is

IDLE → LOADING → ROTATE_TO_UNLOAD
→ DISCHARGE → RETURN_HOME → IDLE.

IDLE admits a box only when automatic operation is enabled, the turntable is at the loading position, and admission is clear. LOADING transfers the box until loaded-position feedback is confirmed. ROTATE_TO_UNLOAD requests the discharge orientation and waits for position feedback. DISCHARGE transfers the box onto its selected, available branch. Confirmed transfer permits RETURN_HOME; loading-position feedback then restores IDLE.

Classification is captured in a defined detection window and retained through transfer. The sensor truth table must account for actual geometry: a tall box may activate both height sensors, so both signals being true is not inherently an error. Missing or inconsistent classification at the end of the window inhibits discharge and raises a diagnostic. The route is cleared only after completion or controlled recovery.

State transitions use a snapshot of the current state and commit one next state per scan. This prevents cascaded transitions caused by LD network order. Automatic requests and manual jog requests are mutually exclusive. Manual mode uses hold-to-run commands subject to the same motion permissives; it does not advance the automatic sequence.

B. Interlocks and Fault Handling

Start sets the run latch only with valid permissives. Stop, a simulated emergency-stop request, or a latched fault removes run permission. Opposing transfer commands are mutually exclusive. Loading and discharge require the corresponding turntable position, rotation requires a valid load condition, and discharge requires the selected branch to be available. Output logic must respect whether a scene tag represents motion or a maintained position command.

Supervision covers entry and exit jams, loading, rotation, discharge and return timeouts, invalid classification, and communication loss. Timers are enabled only during the condition they supervise and reset when that condition ends. Presets must be established from measured travel times with documented margins. A new fault latches its diagnostic and increments the fault counter once. Reset requires the initiating cause to be cleared and does not restart motion; a separate Start and a position/occupancy recovery check are required.

Completed production is counted once per confirmed exit event on the selected branch. Low and high counts must sum to the total count. Interrupted transfers require reconciliation before reset to prevent duplicates. The emergency-stop behavior is a simulation-level functional requirement; ordinary PLC logic and Modbus do not establish a safety-rated stopping function.

IV. Factory I/O and Modbus Integration

The proposed connection follows the vendor's CODESYS setup: Factory I/O operates as a Modbus TCP client and the CODESYS runtime exposes a Modbus TCP Slave Device (server) [2]. The documented baseline uses CODESYS V3.5 SP16 or higher and a Factory I/O edition supporting Modbus. The vendor example uses Structured Text; this project's control strategy is specified in LD.

The commissioning I/O schedule must record every scene tag, PLC variable, direction, data type, bit area, and address offset. Sensor values are written to server coils; actuator commands are exposed through server discrete inputs for the client to read. Verify the selected device mapping rather than inferring direction from the word "input." Additional operator signals need explicit mapping or PLC visualization bindings. IP address, TCP port, unit identifier, channel sizes, and address-base conventions must agree at both ends.

Before automatic operation, toggle each sensor independently and jog each actuator under supervision to verify polarity, address alignment, and direction. Record the PLC task period and communication update interval. Connection status alone does not prove fresh process data: configure a supported watchdog or explicit freshness mechanism, then test cable/driver interruption. Define and verify the simulator's output behavior on disconnect, since a PLC-side inhibit cannot deliver a new command over a broken link.

V. Verification and Acceptance Plan

Status: runtime verification pending. No pass rate, throughput, stopping time, or endurance result is claimed. The checks in Table I are proposed acceptance criteria, not observed results. First review LD network ordering, single-state operation, timer reset paths, output ownership, and the I/O schedule. Then compile the PLC project and record any diagnostics before online testing.

TABLE I
Proposed Acceptance Tests (All Pending)

Test group	Required observation
I/O and modes	Each tag maps correctly; manual and automatic requests cannot overlap.
Normal routing	Low exits left, high exits right; exactly one count per completed box.
Stop and restart	Test every state; commands follow the defined stop policy and no automatic restart occurs.
Jams and timeouts	Block entry, each exit, and each timed transition; correct diagnostic latches and transfer is inhibited.
Classification	Exercise valid tall/short patterns and absent/inconsistent detection; no unclassified discharge.
Reset and recovery	Active causes prevent reset; cleared faults require recovery checks and a new Start.
Communication loss	Interrupt and restore the link; freshness supervision and disconnect output behavior meet the specified policy.
50-cycle run	After functional checks, run 50 mixed boxes with zero misroutes, lost/duplicate counts, or unexplained faults.

For each case, retain the software/scene versions, project revision, initial state, stimulus, expected response, timestamped state and I/O trace, actual response, and pass/fail disposition. During the mixed-box run, reconcile emitted boxes, confirmed exits, and any remaining work in process. Record cycle times and maximum observed fault-detection delay without treating simulation timing as physical-machine performance. Acceptance requires all planned cases to pass and all discrepancies to be resolved or explicitly documented.

VI. Conclusion

The design provides a concise basis for an LD-controlled virtual sorting cell with explicit sequencing, interlocks, diagnostics, and Modbus integration. Its portfolio value rests on traceable control decisions and test evidence. The next milestone is to compile the PLC implementation, validate the complete I/O mapping, and execute the acceptance plan before claiming operational performance.

References

- [1] Factory I/O, "Sorting by Height (Advanced)." [Online]. Available: <https://docs.factoryio.com/manual/scenes/sorting-by-height-advanced/>. Accessed: Sep. 6, 2026.
- [2] Factory I/O, "Setting up CODESYS Modbus TCP (SP16 or higher)." [Online]. Available: <https://docs.factoryio.com/tutorials/codesys/setting-up/codesys-mb-sp16/>. Accessed: Sep. 6, 2026.